Examen LFA

1. Fie $N = (\{0,1\}, \{a,b\}, \{(0,a,0),(0,b,0),(0,b,1),(1,a,1),(1,a,0)\}, 0, \{1\})$. Scrieti o expresie regulata E astfel incat $L(E) = L(N)$. (Bonus: gasiti o expresie regulata compacta)

   Solutie: $a^*b(a \cup b)^*$

2. Scrieti o gramatica G in Forma Normala Cholmsky, astfel incat $L(G) = L(N)$.

   Solutie: $S \to ZS \mid US \mid UA \mid b$, $A \to ZA \mid a \mid ZS$, $Z \to a$, $U \to b$

3. Demonstrati folosind Lema de Pompare ca limbajul $L = \{w_1w_2 \in \{0,1\}^* \mid |w_1| = |w_2| \text{ si } flip(w_1) = flip(w_2)\}$ nu este regulat. $flip(w)$ desemneaza numarul de **alternari** de 0 si 1. De exemplu $flip(01) = 1, flip(10) = 0, flip(00001) = 1$ si $flip(01011100010) = 3$.

   Solutie: $w_n = 0^n1^n0^n1^n$ *si* $i = 2n+1$. *Atunci* $xy^iz = 0^{n+(i-1)b}1^n0^n1^n$. *Pentru* $i = 2n+1, w_n = 0^{n(2+b)}1^n0^n1^n$, *cu* $b \neq 0$. *Asadar* w_1 *va contine* ***doar*** *zero-uri* ($flip(w_1) = 0$) *iar* $flip(w_2) = 2$

4. Scrieti o Masina Turing care accepta limbajul $\{a^nb^n \mid n \geq 1\}$.

   Solutie:

	a	b	A	B	$\#$
0	$1, A/R$			$0, B/R$	$3, \#/H$
1	$1, a/R$	$2, B/L$		$1, B/R$	
2	$2, a/L$		$0, A/R$	$2, B/L$	
3					

5. Aratati ca implicatia: $''\forall L : L^*$ regulat $\implies L$ regulat $''$ este falsa.

   Solutie: $L = \{0^m1^n0^n \mid m,n \geq 0\} \cup \{0\}$. $L^* = L((0^*1^*)^*)$.

6. Aratati ca gramatica: $S \to AS \mid S1 \mid 0S1 \mid \epsilon$, $A \to 0A \mid 0$ este ambigua.

   Solutie: $S \Rightarrow AS \Rightarrow 0S \Rightarrow 0S1 \Rightarrow 01$ *si* $S \Rightarrow 0S1 \Rightarrow 01$ *sunt doua derivari stanga care conduc la constructia a doi arbori de parsare diferiti.*

7. Definiti un APD care accepta limbajul $\{a^nb^m \mid n,m \geq 0, n \neq m\}$

   Solutie: (starile de acceptare sunt 0 si 2)

	a	b	ϵ
0	$0, \epsilon/A$		
1		$1, A/\epsilon$ si $2, Z0/Z0$	$2, A/\epsilon$
2			